

# 18.06 Linear Algebra - Lectures 31, 32  
Textbook Problems - Sections 8.2 & 8.3

° Section 8.2 Problem Set 3, 5-9, 19, 27, 32

3)  $A_1 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$   $A_1^2 \neq A_2$

$\Rightarrow A_1 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

4x4

$A_1^2 : (\cancel{4 \times 4})(4 \times 4)$   
: (4x4).

Want output basis = input basis. ✓

$(A)^2 = B$  when  $(T)^2 = S$  &  
output basis = input basis

5)  $T, A$  Bases:  $v_1, v_2, v_3$   
 $w_1, w_2, w_3$

$T(v_1) = w_2$

$T(v_2) = T(v_3) = w_1 + w_3$

$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$

$A^?$

$\begin{matrix} w_m & v_n \\ w_3 & v_3 \end{matrix} \begin{matrix} (m \times n) \\ (3 \times 3) \end{matrix}$

$A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$  ✓

$T(v_1 + v_2 + v_3) = T(v_1) + T(v_2) + T(v_3)$   
 $= w_2 + (w_1 + w_3) + (w_1 + w_3)$   
 $= \cancel{2w_1} + w_2 + w_2 + 2(w_1 + w_3)$  ✓



6)  $T(v_2) = T(v_3)$   $T(v) = 0$  are  $v = c(v_2 - v_3)$ .

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad Av_0 = 0$$

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} x = 0$$

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} x = 0$$

$$x_2 + x_3 = 0$$

$$x_1 = 0$$

$$\text{Set } x_2 = 1 \text{ so}$$

$$(1) + x_3 = 0$$

$$x_3 = -1$$

$$x_0 = \begin{bmatrix} 0 \\ c \\ -1 \end{bmatrix} \quad \begin{bmatrix} 0 \\ c \\ -c \end{bmatrix}$$

$$v_0 = \begin{bmatrix} d \\ c \\ -d \end{bmatrix}$$

$$T(v) = w_2 \text{ so } v = v_1 \text{ x}$$

all solutions so  $v = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ c \\ -c \end{pmatrix}$

7)  $v \notin C(A)$   $w$  outside  $T(v)$

$$v = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \notin C(A)$$

x  $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$$w_2 - w_3$$

x  $w_1$

$C(A)$  matches range of  $T$ .

8)  $T^2 \neq$  have matrix  $A^2 \rightarrow T(v) \rightarrow T(T(v))$

$$T(v) = w$$

Additional info to confirm:  $T(T(v)) = T(w)$

Do we know  $T(w)$ ?

↳ No

↳ Need to know output

basis matches input

basis.

→ result of Composition Rule



9) rank(A)?

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad n=m=3 \quad \underline{r=2} \checkmark$$

The rank is not the dimension of the whole output space  $W$ .

It is the dimension of the range ~~bases~~ of  $T$ .

↳ The outputs  $AV$  (column space) match the outputs  $T(V)$  (the range of  $T$ ).

↳ The "output space"  $W$  is like  $\mathbb{R}^m$ :

↳ it contains all outputs

↳ but may not be filled up.

19)  $(1, 0) \xrightarrow{B} (1, 4)$   $B = \begin{bmatrix} 1 & 0 \\ 4 & 5 \end{bmatrix}$   $B \begin{bmatrix} x \\ y \end{bmatrix}$   
 $(0, 1) \xrightarrow{B} (1, 5)$

$$\begin{bmatrix} 1 & 0 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

combination:  $a(1, 4) + b(1, 5) = (1, 0)$

$$(a, b) = \underline{(5, -4)} \checkmark$$

$$\begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\Rightarrow a + b = 1$$

$$4a + 5b = 0$$

$$4a = -5b$$

$$b = -\frac{4}{5}a$$

$$a + (-\frac{4}{5}a) = 1$$

$$\frac{1}{5}a = 1$$

$$a = 5$$

$$M = \begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix}; \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 5 \\ -4 \end{bmatrix} = \text{first column of } M^{-1}$$

= coordinates of  $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$  in basis  $\begin{bmatrix} 1 \\ 4 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \end{bmatrix}$ .



32)  $A: (3 \times 4) \quad r = 2 \quad T(v) = Av$

input basis:  $V_1, V_2 \in C(A^T)$   
 $V_3, V_4 \in N(A)$

$$\begin{cases} m = 3 \\ n = 4 \\ r = 2 \end{cases}$$

output basis:  $w_1 = AV_1$   
 $w_2 = AV_2$   
 $w_3 \in N(A^T)$

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

No trick

→ Pretty straight forward

$$r = 2$$

$$(3 \times 4)$$

simple → 1 to 1 for  $w$  when  $v$



3, 4, 15, 17, 22  
 • Section 7.4 Problem Set (Lec 33): Q 4, 5, 16, 17

3) Find  $Q = UV^T$   $\sigma_1 = \sqrt{50}$   $\lambda = 50, 0$   
 $S = V\Sigma V^T$   $V_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$   
 $V_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

$$\Sigma = \begin{bmatrix} \sqrt{50} & 0 \\ 0 & 0 \end{bmatrix} \quad V = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \quad U_1 = \frac{AV_1}{\sigma_1} = \frac{\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \left(\frac{1}{\sqrt{5}}\right)}{\sqrt{50}}$$

$$= \frac{\begin{bmatrix} 5 \\ 15 \end{bmatrix} \left(\frac{1}{\sqrt{5}}\right)}{\sqrt{50}} = \frac{\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}}{\sqrt{10}}$$

$$A = U\Sigma V^T \quad = \frac{1}{\sqrt{50}} \begin{bmatrix} 2 \\ 6 \end{bmatrix} = \frac{1}{\sqrt{10}} \begin{bmatrix} 5 \\ 15 \end{bmatrix}$$

$m \times n \quad m \times n \quad n \times m \quad m \times n$

~~$U_2 = \frac{AV_2}{\sigma_2}$~~   $i = 2 > r.$

$$\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} = U \begin{bmatrix} \sqrt{50} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} = U \begin{bmatrix} \sqrt{10} & 2\sqrt{10} \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} \begin{bmatrix} \sqrt{10} & 2\sqrt{10} \\ 0 & 0 \end{bmatrix}$$

$$U = \begin{bmatrix} \phantom{0} \\ \phantom{0} \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \begin{bmatrix} 2 & 6 \end{bmatrix}$$

$\therefore U_1 = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$   $\therefore U_1 \cdot U_2 = 0$   
 $\therefore U_2 = \frac{1}{\sqrt{10}} \begin{bmatrix} 3 \\ -1 \end{bmatrix}$

$$Q = UV^T = \left(\frac{1}{\sqrt{10}} \begin{bmatrix} 1 & 3 \\ 3 & -1 \end{bmatrix}\right) \cdot \left(\frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}\right)$$

$$= \left(\frac{1}{\sqrt{50}}\right) \begin{bmatrix} 7 & -1 \\ 1 & 7 \end{bmatrix}$$

$$S = V\Sigma V^T = \left(\frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}\right) \begin{bmatrix} \sqrt{50} & 0 \\ 0 & 0 \end{bmatrix} \left(\frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}\right)$$

$$= \frac{1}{5} \begin{bmatrix} \sqrt{50} & 0 \\ 50 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$$

$$= \left(\frac{1}{5}\right) \begin{bmatrix} \sqrt{50} & 50 \\ 50 & 0 \end{bmatrix} = \begin{bmatrix} \sqrt{2} & 10 \\ 10 & 0 \end{bmatrix}$$



3 con.)  $S = V \Sigma V^T$

memo walkthrough

$$\begin{aligned}
 & \left( \frac{1}{\sqrt{5}} \right) \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} \sqrt{50} & 0 \\ 0 & 0 \end{bmatrix} \left( \frac{1}{\sqrt{5}} \right) \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \\
 &= \left( \frac{1}{\sqrt{5}} \right)^2 \begin{bmatrix} \sqrt{50} & 0 \\ 2\sqrt{50} & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \\
 &= \left( \frac{1}{5} \right) \begin{bmatrix} \sqrt{50} & 2\sqrt{50} \\ 2\sqrt{50} & 4\sqrt{50} \end{bmatrix} \\
 &= \left( \frac{1}{\sqrt{50}} \right) \left( \frac{1}{5} \right) \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} (\sqrt{50}) \\
 &= \sqrt{2} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \\
 &= \frac{1}{\sqrt{50}} \begin{bmatrix} 10 & 20 \\ 20 & 40 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 A &= QS \\
 &= \left( \frac{1}{\sqrt{50}} \right) \begin{bmatrix} 7 & -1 \\ 1 & 7 \end{bmatrix} \left( \frac{1}{\sqrt{50}} \right) \begin{bmatrix} 10 & 20 \\ 20 & 40 \end{bmatrix} \leftarrow \odot
 \end{aligned}$$

$S$  is semidefinite because  $A$  is singular.

4)  $A^+ = V \Sigma^+ U^T$        $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$

$$\begin{aligned}
 &= \left( \frac{1}{\sqrt{5}} \right) \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{50}} & 0 \\ 0 & 0 \end{bmatrix} \left( \frac{1}{\sqrt{5}} \right) \begin{bmatrix} 1 & 3 \\ 3 & -1 \end{bmatrix} \\
 &= \left( \frac{1}{5} \right) \begin{bmatrix} \frac{1}{\sqrt{50}} & 0 \\ \frac{2}{\sqrt{50}} & 0 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 3 & -1 \end{bmatrix} \\
 &= \left( \frac{1}{5} \right) \begin{bmatrix} \frac{1}{\sqrt{50}} & \frac{3}{\sqrt{50}} \\ \frac{2}{\sqrt{50}} & \frac{6}{\sqrt{50}} \end{bmatrix} = \left( \frac{1}{5} \right) \left( \frac{1}{\sqrt{50}} \right) \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \\
 &= \left( \frac{1}{5} \right) \left( \frac{\sqrt{2}}{10} \right) \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \\
 &= \frac{\sqrt{2}}{50} \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \quad \times
 \end{aligned}$$

$$\begin{aligned}
 A^+ A &= \left[ \frac{1}{50} \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \right] \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} = \left[ \frac{1}{50} \right] \begin{bmatrix} 10 & 20 \\ 20 & 40 \end{bmatrix} \\
 &= \begin{bmatrix} 0.2 & 0.4 \\ 0.4 & 0.8 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 A A^+ &= \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \left( \frac{1}{50} \right) \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} = \left( \frac{1}{50} \right) \begin{bmatrix} 5 & 15 \\ 15 & 45 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix} \\
 &= \begin{bmatrix} 0.1 & 0.3 \\ 0.3 & 0.9 \end{bmatrix}
 \end{aligned}$$



15)  $A, AA^T, A^T A, A^+ \rightarrow$  all rank = 1

a)  $A^T A \hat{x} = A^T b$  many solutions  $\Rightarrow A^T A$  is invertible <sup>singular</sup>  
 $\hookrightarrow$  due to full row rank.

b)  $x^+ = A^+ b = (0.2b_1 + 0.1b_2, 0.2b_1 + 0.1b_2)$

$$A^T A \hat{x} = A^T b$$

$$\begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix} \begin{bmatrix} 0.2b_1 + 0.1b_2 \\ 0.2b_1 + 0.1b_2 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$= \begin{bmatrix} b_1 + 0.5b_2 \\ b_1 + 0.5b_2 \end{bmatrix} = \begin{bmatrix} 2b_1 + b_2 \\ 2b_1 + b_2 \end{bmatrix}$$

$$= \begin{bmatrix} 2b_1 + b_2 \\ 2b_1 + b_2 \end{bmatrix} \Leftrightarrow$$

c) ?  $(1, -1) \rightarrow$  in  $N(A)$

So if add (its orthogonal to  $C(A^T)$ ) to  $\hat{x}$   
 the  $\hat{x}$  will be longer than  $x^+$ .

$$\|\hat{x}\|^2 = \|x^+\|^2 + \underbrace{\|\text{added } (1, -1) \text{ component}\|^2}_{\text{extra}}$$

17)  $\mathbb{B}$  In  $\mathbb{R}^m$  :  $b = p + e$   $C(A) + N(A^T)$   
 In  $\mathbb{R}^n$  :  $x = x^+ + x_n$   $C(A^T) + N(A)$

$\circ \circ AA^+p =$   ~~$AA^+p$~~

~~$(m \times n)(n \times m) \Rightarrow (m \times m)$~~

(input) (col space)  $\rightarrow$  col space  $A^+$  is a projection  
 $AA^+p = p$  ✓

$A^+$  is a projection

$AA^+e = ?$

$AA^+e = 0$  ✓

$\hookrightarrow AA^+$  project onto  $C(A)$

input into <sup>left</sup> null space = 0.

$A^+A x^+ = ? \Rightarrow x^+$  ✓

$\hookrightarrow A^+A$  project onto  $C(A^T)$

$A^+A x_n = ? \Rightarrow x_n = 0$  ✓